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1 Introduction (2-3pg)

In electrophysiology, two prevalent modes of biopotential measurements are electrocardiography (ECG) and electromyography (EMG), measuring cardiac muscle potentials and skeletal muscle potentials respectively. While methods that use intracellular electrodes exist and are much more precise, using surface electrodes is very common thanks to their non invasive nature, albeit at a loss of spatial specificity due to them being limited to measuring a superposition of nearby potentials. The field of ECG is by and large applied to identifying and diagnosing cardiac abnormalities in clinical and first responder settings, focusing on disturbances of the cardiac rhythm, artery blood flow, or electrolyte availability for diagnosis of heart conditions. On the other hand, Surface EMG (henceforth sEMG) has broader applications, ranging from the clinical, such as diagnosing nerve and muscle damage, to prosthetics, rehabilitation, biofeedback therapy, ergonomics, sports and movement science [1], and it also opens up the possibility of novel Human-Computer Interaction methods.

Robot Operating System (ROS) is an open-source software framework for robotic applications, most importantly providing a flexible communication system ("the plumbing") that allows for real-time message-passing between networked nodes. In practice, these nodes can be any component of the larger robotic system, be them anything from a discrete piece of hardware to a software component. Nodes communicate by sending messages, which can have varying levels of complexity, from primitives to nested aggregated structures, via "topics" which implement publish/subscribe semantics. In our application, the biopotential DAQ node can expose topics on which it publishes measured samples, for any other node to subscribe to for further signal processing.

The Linux Industrial I/O (IIO) subsystem was created in 2008 and mainlined in 2012 (commit 06458e2) to provide a simple and unified way to interact with devices with ADC or DAC capabilities, broadly ranging from plain ADC/DAC chips to Inertial Measurement Units (Accelerometers, Gyroscopes, Magnetometers), Software Defined Radio, robotic actuators and sensors, and more. Within the Linux kernel, it fills in a gap between the *input* subsystem which can stream user input data and the *hwmon* subsystem used for monitoring and controlling hardware state such as CPU temperature and fan settings, albeit with low refresh rate. The IIO subsystem provides a modular way to define high bandwidth communication with input/output devices directly connected to the host via SPI or I2C and specified as part of the device tree. Despite the subsystem's name, it is not limited to industrial production applications: IIO usage is ubiquitous in today's linux-based devices for integrating features such as accelerometers, gyroscopes or ambient light sensors in a uniform way, besides other hardware abstraction layers.

Libiio is a library which provides cross-platform functionality for interacting with both local and remote IIO devices within the same framework. To accomplish this, it implements the IIO Daemon (IIO Daemon) for which it defines the *IIO Daemon protocol*. The current version of the IIO Daemon protocol (v0.1) is based on a set of human readable commands sent via a single data stream of alternating client requests and server responses. This allows the same protocol to be used over a variety of transport layers, officially Serial (most relevant for this application), USB and TCP.

Although the electrical design of biopotential measurement devices has been studied for hun-

dreds of years, tracing back to Luigi Galvani’s work in the 18th century, and there exist a number of current-day commercial, academic, and DIY solutions, only a small part aspire to the ideals of Open Source Hardware and/or Open Source Software (ASPEN[2], openBCI[3], BioAmp-EXG-Pill[4], MyoWare EMG[5]), and there exist no systems that integrate with ROS2 or the Linux IIO subsystem, limiting their flexibility in terms of building software that seamlessly interfaces with them.

We aim to fill this gap and provide an affordable and easily maintainable hardware and software solution for measuring multi-channel biopotential signals and interfacing them with ROS2 and LibIIO. Adapting this system to work with other software suites should be made as simple as possible thanks to the versatility of both ROS2 and LibIIO, and adapting it to other hardware platforms should also be facilitated by the decoupled nature of the hardware, firmware and software parts. An overview of the information flow, from the biological signals to the software, in figure 1.1.

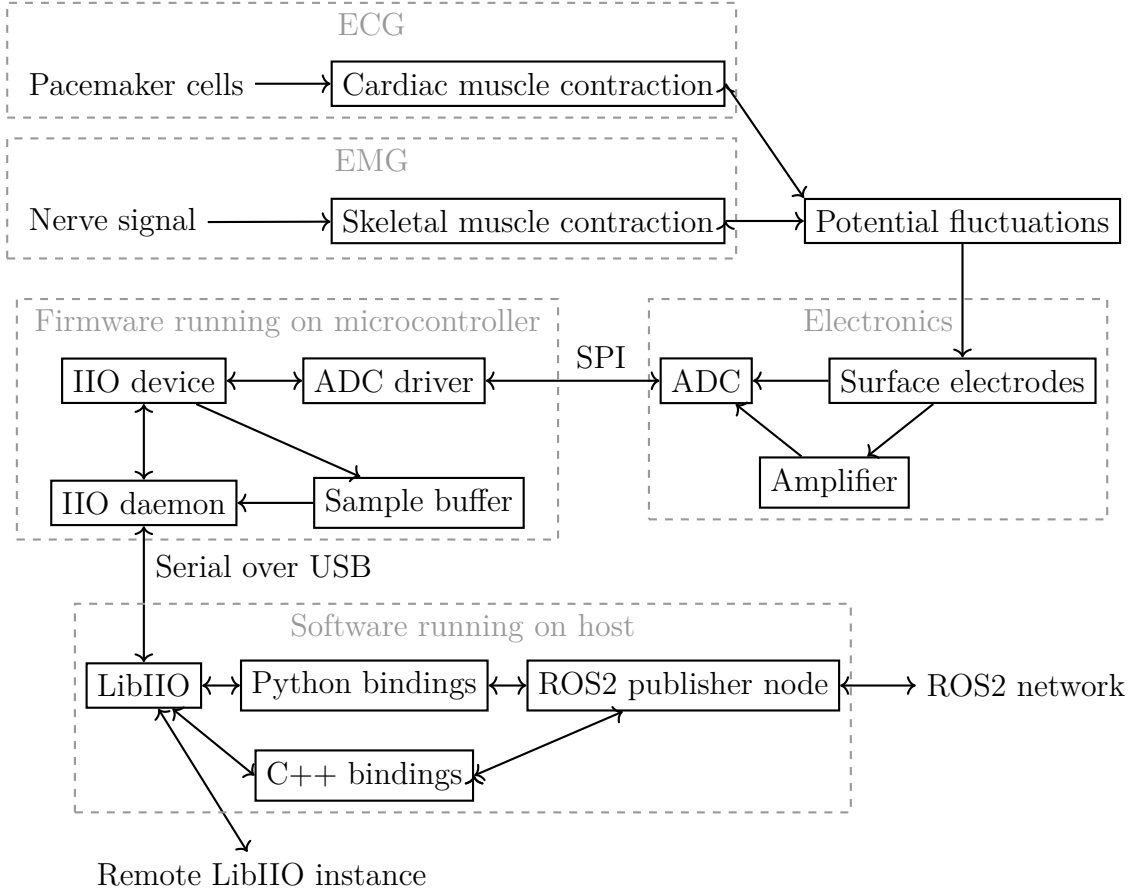


Figure 1.1. Overview of information flow

2 Bibliographic research (15pg)

2.1 Anatomical basis

As part of normal function, all biological cells have a difference of electrical potential between their interior and their exterior, called the *membrane potential*. In the steady state, the *resting membrane potential* of both skeletal muscle cells and that of cardiomyocytes (cardiac muscle cells) is in the -70 to -90mV range [6], [7].

2.1.1 Skeletal muscles: sEMG

From an electrophysiological standpoint, skeletal muscles are comprised of a number of motor units (MU), each corresponding to an individual axon that innervates a number of muscle fibers. Neural impulses travel from the central nervous system via nerves to a motor unit's innervation zone (IZ), where a number of neuromuscular junction synapses trigger an action potential in the fibers of that motor unit. The electrical activation of the muscle doesn't take place all across the fiber simultaneously, but rather travels from the innervation zone towards the tendons as an impulse.

This starts out as an action potential (AP) at the innervation zone, wherein the span of 4-6ms, the membrane potential rises from the resting -70mV to positive (depolarisation) and then back to the resting potential (repolarisation) by biomolecular mechanisms involving the flow of Na^+ , K^+ and Cl^- ions. This localized action potential also triggers action potential in neighbouring parts of the muscle cell, leading to a chain reaction down the muscle fiber with a clear external electrical manifestation: the Motor Unit Action Potential as a traveling impulse of ion flow, which is measurable via the potential fluctuations across tissue.

Muscle fiber contractions follow the "all or nothing" principle, meaning a fiber either contracts fully or not at all. To modulate the force exerted by a muscle, only a part of the available motor units are recruited. Still, there is an independence between the number of motor units, number of fibers, and a muscle's strength, the motor unit count only improving the "resolution" of muscle control. By Henneman's size principle [8], low effort contractions start by recruiting smaller motor units and gradually activate larger motor units as load increases. Besides this spatial control, muscular contraction is also controlled via rate coding, where the same motor unit is reactivated repeatedly, with the activation frequency positively correlating with the overall contraction strength, until fatigue is reached [9].

As a practical result, during low effort motion where only one or a handful of motor units are engaged, the APs are very easily identifiable in the measured waveform. On the other hand, high effort contractions will manifest a superposition of uncorrelated MUAPs which are difficult to separate from just sEMG measurements. (figure 2.1)

Surface measurements have worse spatial resolution and cannot precisely pinpoint the source of the observed electrical activation, but rather "see" a potential halo. In the case of muscle fibers, the surface halo of a travelling action potential is modelled as a directional tripole with a leading positive depolarizing area, a trailing positive repolarizing area, and a strongly negative spike in the middle.[1, Section 3]

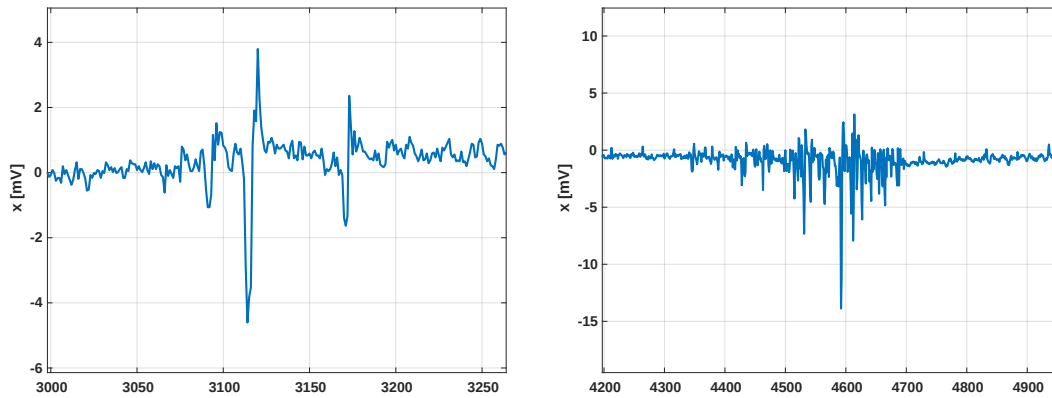


Figure 2.1. Individually identifiable action potentials during a low-effort period (left). Superposition of action potential during a higher-effort movement (right). Electrodes placed 35mm apart on inner forearm.

2.1.2 Cardiac muscles: ECG

While the internal mechanisms of muscle contractions are similar between cardiac and skeletal muscle, their external activation strategies vary almost entirely. Cardiac muscle isn't divided into motor units and as such heartbeat strength cannot be modulated by recruitment, but rather by various chemical and mechanical factors that ultimately increase or decrease the availability of Ca^{2+} ions, such as but not limited to: the Frank-Starling mechanism (which aids in stroke volume regulation), or hormone effects from secretion of adrenaline or norepinephrine.

Contrasted to the usual sEMG measurement, all healthy heartbeats have a clearly identifiable structure consisting of a P wave, the QRS complex, and a T wave. **TODO: FIGURE** Cardiac conditions can be characterized and diagnosed by these features' timing, but their electrical characteristics are very well defined, especially when compared to the noisy-looking MUAP superposition seen during skeletal muscle contraction.

Another main difference between sEMG and ECG is electrode placement. While for sEMG we are interested in getting the electrodes as close as possible to the muscle of interest to avoid confusion with other sources of electrical activity, ECG focuses on large scale electric field gradients that appear in the body as a result of a heartbeat. In essence, both the amplitude and "direction" of depolarization and repolarization can be determined by measuring potential differences across a patient's extremities. Localized contractions of cardiac muscle can be identified by the electric field disturbances they create even tens of centimeters away, bypassing the need to have electrodes as close as possible to the heart. All of this is possible in ECG but not EMG because of the very strong electrical impulses created by the heart, as well as their regular, non-superimposed structure. **TODO: FIGURE illustrating cardiac dipole model**

2.2 Electronics for biopotential measurement

2.2.1 Biopotential measurement modes

Two common modes of measurement of biopotentials arise: *unipolar* measurements referred to a common point whose potential isn't affected too much by the observed process and *bipolar* measurements where consecutive electrodes' potentials are taken into differential amplifiers, as illustrated in figure 2.2. Both are relevant for a biopotential data acquisition system, and as

such one of our goals is having a system that can do both.

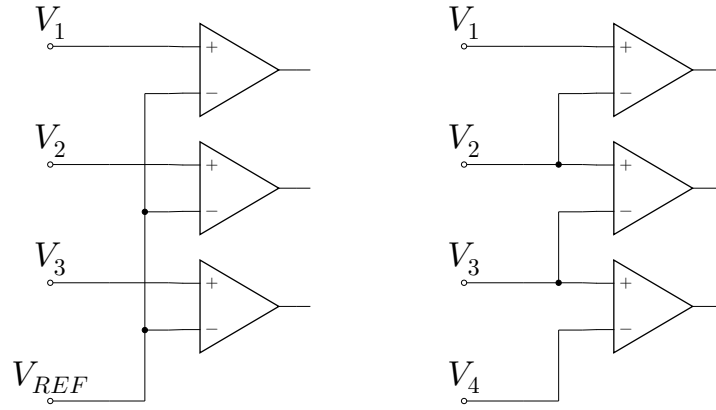


Figure 2.2. Unipolar (left) and bipolar (right) measurements with 4 electrodes

2.2.2 Driven Right Leg

In both measurement modes, having the patient's body properly referred to a known electrical node is very important for the proper functioning of amplification and sensing circuitry. If the patient is "disconnected" from the rest of the circuit, and only interfaced to with high impedance sensing electrodes, stray charges originating from various sources can push the entire patient's body to common-mode potentials above or below the measurable range. Static electricity from clothing can regularly build up hundreds or thousands of volts, which easily exceeds the measurement range of common ADCs or amplifiers. While electrostatic discharge can be accounted for with protection circuitry, such high common-mode voltages make signals unrecoverable.

The solution is having a current path from the observed patient to the measurement apparatus' ground, through which charge imbalances can be equalized. This can just be a passive connection to ground (figure ?? a), but using active components on this path allows us to modulate the potential difference between the patient and system ground, which opens up the opportunity to both bias all signals to a known voltage, as well as cancelling predictable or low-frequency noise, such as mains hum, by feeding its opposite as a bias.

For the practicality of such a biasing circuit, it must connect to the patient in a relatively far away part of the body from what is being measured, so it will minimally interfere with the other electrodes, with as little introduced differential voltages as possible. As per ECG doctrine, the right leg is often used, leading to the name "Driven Right Leg circuit", but this referencing electrode can be connected anywhere depending on the measurement at hand.

2.2.3 Electrode types

The first step we control in the signal chain is the point of contact with the patient. The electrode type and quality have a great influence on the overall design, performance, and subjective user experience.

In the simplest case, just a metallic electrode connected to the body can sense electrical field changes, forming a polarizable electrode, which cannot offer consistent current flow, but rather acts like a capacitor, wherein ion concentrations on the skin surface will repel or attract electrons in the electrode, creating small currents.

A much more advantageous setup is one in which the physiological ion currents can be directly converted to electrical current akin to a resistive interface, termed a non-polarizable electrode. By far the most commonly used electrodes are Ag/AgCl electrodes which fit into this category.[10] These have a silver chloride surface layer, backed by a silver core, and historically other metals for cost reduction. These are commonly distributed as both dry electrodes, which contact the skin directly, but require an electrolyte solution to be applied for best results, as well as wet electrodes, with the electrolyte solution suspended in an elastic gel that adheres to both the electrode and skin, both illustrated in figure ???. The electrolyte solution provides free Cl^- ions that facilitate the conversion of electric current to ionic current by ionic decomposition of the AgCl layer or recombining Cl^- ions with Ag^+ ions. Dry electrodes can also somewhat function without any electrolyte solution applied thanks to the dissolved NaCl content of sweat, which also provides free chlorine cations.

Chemically, the AgCl layer partially decomposes into Ag^+ and Cl^- ions when exposed to the aqueous electrolyte solution. When the body potential is higher than the electrode potential, i.e. there's a lack of anions in the body, chlorine cations flow from the electrolyte gel towards the body, further encouraging the decomposition reaction $\text{AgCl} \rightarrow \text{Ag}^+ + \text{Cl}^-$ to replenish them. The positive Ag^+ can precipitate out on the electrode surface, exchanging an electron and thus inducing conventional electrical current "into" the electrode. On the other hand, when the body is more negatively charged than the electrode, Cl^- ions in solution will be in excess and will recombine with Ag^+ ions on the AgCl surface. The new lack of silver anions promotes the ionization of the metallic silver layer, releasing electrons. In both cases, the Ag/AgCl electrode and electrolyte solution act like a mostly resistive interface, with some stray parallel capacitance.

Because of their prevalence, Ag/AgCl electrodes are inexpensive and easily obtainable from general medical suppliers. Additionally, [11] presents 3 methods for fabricating dry Ag/AgCl electrodes, which can be implemented even in small scales and using cheap materials, were these not commercially available.

The state of the art are conductive polymer PEDOT:PSS electrodes, thanks to their great mechanical, electrical, and chemical properties: They adhere and conform to skin, are flexible, self-healing, and have good biocompatibility, only falling behind the classic Ag/AgCl electrodes in terms of contact impedance, yet still being conductive enough for biopotential readings. [12], [13]

2.3 Market research

Biopotential measurements are no new subject and many products already exist both in the academic and commercial fields. All are subject to engineering tradeoffs between parameters such as number of channels, sample rate, precision, noise, electrode types, etc., making each better suited to a certain use case. To the best of our research, there is no universal, modular and extensible solution to the problem of biopotential measurements. Further on, we overview some representative options for EMG / ECG DAQ systems:

OpenBCI

The OpenBCI ("Open Brain-Computer Interface") project started in 2013 and provides open source boards mainly intended for EEG measurements, but which are also applicable for EMG and ECG. These have undergone a handful of revisions in the last decade, and the current generation consists of:

- Cyton – an 8 channel, 24 bit, DAQ based on the ADS1299. Retail for 917 Euro (as of 2024-08).
- Ganglion – a 4 channel, 24 bit, DAQ based on the MCP3912. Retail for 458 Euro (as of 2024-08).
- Daisy – an add-on for the Cython providing 8 extra channels using a second ADS1299. Cython+Daisy package retail for 1835 Euro (as of 2024-08).

While the retail price on the OpenBCI store is steep, design files are provided for all products and they can be ordered and individually built for a fraction of the cost. If built based on the schematics, the bulk of the cost will be the ICs, which add up to around 100 Euro (estimate as of 2024-08).

ASPEN

[2]

What we're doing, but better, but a PhD thesis.

TODO: more :)

Quanser QNET Myoelectric board

The Quanser QNET Myoelectric trainer board is a closed-source, one channel add-on board for the National Instruments Elvis II. It provides a fixed amplification stage of powered by a battery to remove power supply noise, and further uses the ELVIS' oscilloscope channel for analog to digital conversion. Its signal chain consists of an overall gain of 300, and first order LPF and HPF limiting the bandwidth to 20 – 500Hz.

Anecdotally, this is the board that sparked the idea for this project, as it is one of the experimental boards given to TUCN students. It depends on the NI Elvis, only has one channel measurements, and is not open source. This makes it a good educational resource, but it falls short on both performance and cost.

BioWare EXG Pills

As one of the cheapest related products, the EXG Pill is an open source instrumentation amplifier board meant to aid in biopotential data acquisition by connecting it to another device with an ADC, typically an Arduino board. It is available for order on the creator's online shop, but being an open hardware project allows anyone to manufacture their own, to specification. It is not an entire DAQ system, but rather just the amplification stage.

Cometa Pico EMG, Delsys Tringo

As one of the most polished products on the market, the Cometa Pico EMG and Delsys Tringo are two competing multi-channel wireless EMG DAQ systems.

The Cometa Pico EMG supports up to 36 channels. Its sensors have a 2kHz sample rate and an integrated IMU.[14]

The Delsys Tringo has a more flexible system with a range of different sensor types: single bipolar sensors, grid array sensors, snap electrode sensors, which are more intended for high end research. Comparable to the Pico EMG, the Trigno Avanti sensor has a max sample rate of 4370 sps and also have integrated IMUs.[15]

On the other hand, being high-end commercial medical products, both systems cost orders of magnitude more than what we aim for ([16] estimates 15000 Euro for the Cometa and 20000 Euro for the Delsys) and neither abide by the open source philosophy.

MyoWare

Advancer Technologies's line of MyoWare products started out in 2015 with their first generation MyoWare muscle sensor, which similarly to the BioWare EXG Pills described above, only provided the analog signal chain consisting of isolation, an adjustable amplifier and filtering, requiring an external ADC, with the Arduino Uno being recommended for small projects. The current generation has adjustable gain between ? — 200 and a bandpass filter between 20 — 500Hz.[5]

2.4 Noise sources

The superposition of electrical signals measurable by biopotential DAQs contains a complicated mix of noise coming from various physiological and other external factors, besides the signal of interest. Additionally, the measurement circuitry compounds noise at every step. Modeling some of the main noise components of the entire signal chain is essential for its design and proper specification.

Some of the studied noise sources are *intrinsic* to the process and impossible to remove, such as thermal noise and the amplifier and ADC noise profiles. Others are not intrinsic, depend on the setup, and can be compensated for, such as mains hum.

2.4.1 Thermal noise

Even with ideal conductors and measurements, any resistive circuit will be affected by thermal noise, as a consequence of Planck's law of blackbody radiation [17, chapter 3.2]. An ideal resistor R at absolute temperature T will manifest thermal noise V_n over a frequency window Δf having mean squared value $\overline{V_n^2} = 4k_B T R \Delta f$, where $k_B = 1.380649 \cdot 10^{-23}$ J/K is the Boltzmann constant.

For a rough estimate of the order of magnitude of thermal noise in our biopotential measurements, we study the following hypothetical setup: Assume the electrodes attached to the patient are in thermal equilibrium at a typical skin temperature of $T \approx 308$ K [18], we are limiting the frequency band to $\Delta f = 500$ Hz by filtering, and experiencing a typical resistance

$R \approx 3500\Omega$ (electrode + contact + internal resistance) [19]. This is only useful as a rough order estimation, because the resistance will vary depending on the electrode location, surface preparation, measured individual, electrode condition, etc. The estimated thermal noise will be:

$$\overline{V_n^2} = 4k_B T R \Delta f \approx 0.02977 \mu V^2 \quad (2.1)$$

Which can alternatively be interpreted as having an RMS noise of $\sqrt{0.02977 \mu V^2} \approx 172.5 \text{ nV}$ or a noise density of $\sqrt{4k_B T R} = 7.716 \text{ nV}/\sqrt{\text{Hz}}$.

2.4.2 Amplifier and ADC noise

Both ADCs and amplifier circuits have specified noise figures which pose design tradeoffs. In ADCs, the noise is specified as additive noise at the input V_{na} , such that a measurement V_{meas} of a "real" voltage V will be $V_{meas} = V + V_{na}$. Amplifiers are characterized by both input noise V_{ni} created by the input stage, as well as output noise V_{no} created by internal circuitry and the output stage. The output of an amplifier with gain G is $V_{amp} = (V + V_{ni}) \cdot G + V_{no}$. As an easier to use figure, the *Input Referred Noise* is the equivalent disturbance at the input that would create both the effects of input and output noise compounded. $V_n = V_{ni} + V_{no}/G$ such that we can simplify to $V_{amp} = (V + V_n) \cdot G$. An equivalent block diagram is presented in figure 2.3.

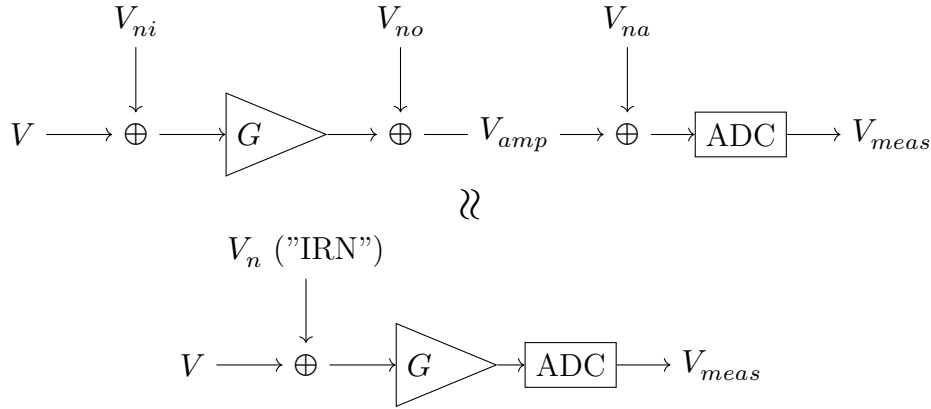


Figure 2.3. Block diagram of amplifier and ADC signal chain, with modelled noise

This raises one of the most important design decisions of the signal chain, namely deciding on the amplification structure before the ADC. Using no amplifier at all is the most economic and also avoids V_{ni} and V_{no} , but in the case of very small input signals, they are left to be dominated by ADC noise V_{na} . Adding an amplifier with unity gain only compounds amplifier noise and is as such detrimental, but further increasing the gain leads to better signal-to-noise ratio (SNR):

$$\text{SNR} = \frac{\text{"signal"}}{\text{"noise"}} = \frac{G \cdot V}{G \cdot V_{ni} + V_{no} + V_{na}} = \frac{V}{V_{ni} + (V_{no} + V_{na})/G} \rightarrow \frac{V}{V_{ni}} \quad (2.2)$$

Thus, given an amplifier with known V_{ni} we can compute the best-case noise figure assuming "infinite" gain. The rationally convergent nature of the above expression inevitably creates a

situation of diminishing returns, wherein increasing the gain will always increase the signal quality, but even relatively small gains achieve practically ideal performance.

The drawback of having very large gain is the limit it incurs on the measurable range of input voltages. Any analog circuit can only handle a limited interval of voltages, which when exceeded leads to unreliable measurements, with the possibility of permanent damage to the equipment. If the ADC is limited to measuring $\pm V_{max}$, the measurable signals are implicitly limited to $\pm V_{max}/G$.

In this topology consisting of a single amplifier stage characterized by V_{ni} and V_{no} , followed by an ADC with input noise V_{na} , measuring the "true" voltage V , the resulting digitized sample V_{meas} can be decomposed into:

$$V_{meas} = GV + GV_{ni} + V_{no} + V_{na} \quad (2.3)$$

Which can be rearranged to obtain an expression for RMS total noise V_n :

$$\overline{V^2} = \frac{\overline{V_{meas}^2}}{G^2} + \underbrace{\left(\overline{V_{ni}^2} + \frac{\overline{V_{no}^2} + \overline{V_{na}^2}}{G^2} \right)}_{\overline{V_n^2}} \quad (2.4)$$

2.4.3 Flicker noise

Especially prevalent in BJT transistors and devices built using them, flicker noise is a low-frequency phenomenon where various material defects induce fluctuations with power density $P \propto 1/f$. It is also manifested in other active devices (MOSFETs, JFETs) and passive components, with a mix of various mechanisms contributing to it, many of which not fully agreed upon. A practical yet relatively complex solution to flicker noise is modulating the signal of interest to higher frequencies using a chopper and lock-in amplifier. [20]

2.4.4 ECG noise models

Besides the previous intrinsic kinds of noise, further disturbances in ECG measurements can be classified, as per [21], into: low frequency ($< 0.5\text{Hz}$) baseline wander caused by body motion and electrode contact issues, power-line artifacts (50/60Hz and their harmonics) caused by insufficient power supply decoupling when the measurement device is connected to mains power and capacitive or inductive coupling between the patient and surrounding mains lines, and muscle artifacts (EMG signals). Of the three, the first two are relevant and undesirable in any biopotential measurement, while the latter is obviously the signal of interest in EMG applications.

From a signal processing standpoint, a simple solution for low frequency wander and power line artifacts is using a DC blocking comb filter, but these can also be addressed in the hardware design, especially when it comes to power supply decoupling.

2.5 Signal chain design

With today's available technology, signals in the millivolt range are relatively easy to measure even at quite high rates, if strongly driven. Intrusive methods of biopotential acquisition such as needle electrodes plugged directly into muscle fibers can work with no amplification stage, and just a direct connection to the ADC board, while still having good signal quality. On the other hand, when measuring with surface electrodes, both the electrode and biological tissue become a very high impedance element in the signal chain, which means that even relatively high biopotentials will have a hard time overcoming parasitic effects and signal chain noise before reaching the ADC.

2.5.1 Noise and gain in cascaded signal processing structures

2.5.2 Instrumentation amplifiers

Instrumentation amplifiers are a class of differential amplifier which boast high open-loop gain, low noise, and high input impedance. The typical textbook realization is with 3 operational amplifiers connected like in figure 2.4, which allow the gain to be adjusted by choosing the value of resistor R_G . Their high input impedance means that very little current is needed to affect the inputs, helping with the issue of high electrode and tissue impedance. Their high power supply rejection ratio (PSRR) and common mode rejection ratio (CMRR) make them even more suitable for differential electrode applications, requiring very little extra circuitry.

Figure 2.4. Typical implementation of an instrumentation amplifier

2.5.3 Multiplexed ADCs

TODO: write about them :) - multiple channels, single converter, multiplexor (possible configurations - limited or arbitrary?), digital filtering, sequencer, conversion clock dependence on communication clock makes noise and is bad

2.5.4 EMI prevention over long wires

When dealing with long wires and weak signals, electromagnetic interference (EMI) is a significant source of noise which is produced by virtually all electronic devices around. To combat this, common techniques are using twisted pairs and wire shielding. For simplicity, we will try to shorten the wires carrying weak signals as much as possible and only use strongly driven and high amplitude signals over long wires. The result is that in bipolar measurements, we place the differential amplifier as close as possible to the electrodes, and then don't care about signal degradation between the amplifier and ADC thanks to the strongly driven nature of that signal.

2.5.5 Common collector voltage buffer

Later used in both the preamplifier and Driven Right Leg circuits, the common collector amplifier, illustrated in figure 2.5 is a minimal and cheap circuit for providing a low impedance, high drive strength "mirror" of an incoming signal with potentially higher impedance.

Assuming the input voltage puts the transistor in the linear operating range, the output voltage will

circuit frumos :)

Figure 2.5. Common-collector voltage buffer

3 Analysis, design and implementation (30pg)

3.1 Objectives

Additionally, we

TODO: Justify objectives, format these boring bullet lists in a nicer way

MUST:

- Open Hardware and Software
- 8 channels
- Per-channel configuration
- 1000Hz data rate (500Hz bandwidth) TODO: NNNNNNOOOOOOOOOO
- High CMRR, PSRR
- High SNR (>40dB EMG; >20dB for ECG)
- Open and extensible interfacing: ROS2, IIO

SHOULD (unchecked go to future work):

- 16 channels
- Differential / single-ended configurability
- Right Leg Drive
- 1024Hz/2048Hz /channel data rate (1 Hz per FFT bin)
- <1uV noise after filtering
- 1uV precision

Guiding principles:

- Cost
- Usability as an educational tool
- Use Analog Devices parts
- Modularity (decoupled hardware-firmware-software)

TODO: After finishing the prototype, make a comparison table between the already existing products and mine, especially on these objectives

3.2 Electrical part

3.2.1 Interfacing with the patient

Taking from the research from section ??, we do not consider PEDOT:PSS electrodes to be viable for this project due to them not being commercially available in standard and mass produced forms, being mostly a custom product experimented with in today's laboratories, even half a century after their discovery in the 1980s.

We settle for snap-on adhesive Ag/AgCl gel electrodes thanks to being cheap, readily available, standardized, and they can be connected to with inexpensive alligator clips, besides special (but more expensive) snap clips. The downsides are their low shelf life **TODO: compare expired electrode signal quality with fresh electrodes once I get some fresh ones**, comparatively larger contact area (cca 200mm² vs [1]'s 2mm electrode diameters), and are lacking in subjective user experience, sometimes being very painful to detach.

3.2.2 ADC

We initially chose the AD7124-8[22] thanks to its very diverse feature set, including: programmable gain amplifiers (PGAs) which had the potential to remove the need for the external amplification stage, high precision, high number of channels, and high sample rate. Unfortunately, it is not a single cycle ADC. This means that although it has very good throughput on a single channel, when switching between two channels there is a dead time for the integrated filter to settle, which effectively reduces the per-channel sample rate to less than 100Hz when sampling all 16, which is virtually useless for ECG ($f_{\max} = 150\text{Hz}$) and EMG ($f_{\max} = 500\text{Hz}$).

Our next choice was the AD4114[23]. It doesn't include PGAs, but otherwise has very good specifications and has a much higher sample rate and a single cycle conversion mode.

The AD717X family of ADCs, which the AD4114 is part of (in spite of the numbering scheme), all expose a similar interface to the application developer. A number of channels can be enabled, which are then measured in a loop, in ascending order of their indices. Each of the 16 channels can select one of 34 inputs: 16 unipolar measurements ($V_{in,k} - V_{incom}$), 16 bipolar measurements ($V_{in,2k} - V_{in,2k+1}$ and $V_{in,2k+1} - V_{in,2k}$), an internal temperature sensor or the voltage reference itself. If an input is selected by multiple channels, it is read multiple times in the sequence. To parametrize a channel's conversion, one of 8 "setups" can be assigned. Setups describe whether a measurement is unipolar or bipolar, the used reference, buffering and filtering settings. Each setup has a corresponding digital filter with an internal state. If multiple channels refer to the same setup, they will also be alternately using the same digital filter, requiring some extra logic for avoiding latent (i.e. "hidden") samples from passing from one channel's to another's readouts. Conveniently, the SING_CYC flag puts the ADC in "single cycle mode", which always flushes out the digital filter when switching channels, at the expense of worse filter performance, data rates, and greater noise.

The AD4114 lacks an internal sample buffer, meaning that data from a conversion must be read before the next one or else it is lost. While the digital interface allows for an interrupt signal on the DOUT/ $\overline{\text{RDY}}$ line to notify the host of new data, it is incompatible with most microcontrollers' hardware SPI implementations and it is much more suitable for an FPGA host. Therefore we can and should resort to polling the status register to check for new samples, which are an-

Table 3.1. Selected measurement configurations and their specifications

N	ODR (Hz)	Filter	Per-channel rate (Hz)	Noise ($\mu\text{V RMS}$)
1	1007	sinc5+sinc1	1007	20
2	2597	sinc5+sinc1	1298.5	38
3	3115	sinc5+sinc1	1038.33	41
4	4444	sinc5+sinc1	1111	56
5	5181.65	sinc5+sinc1	1036.33	64
6	6211	sinc5+sinc1	1035.17	74
7+	10309	sinc3	$10309/N$	1068

nounced through the RDY bit. Polling should be done at least as fast as the conversion rate, but SPI transactions are cheap enough to poll even an order of magnitude more often than samples are coming in.

One of the most important limitations in determining which parameters to run the ADC in is its noise, which varies with the sample rate. While high sample rates are very desirable given we have practically limitless digital signal processing capacity, they also degrade the signal quality. This is especially relevant when sampling multiple channels, where the Output Data Rate (ODR) is divided between them. A single channel measurement with $\geq 500\text{Hz}$ bandwidth can be done with the 1007Hz setting for an effective resolution of 17.3 bits, but having 6 channels forces the use of the 6211Hz setting with 15.4 bits of resolution, and more channels than that start limiting the bandwidth to less than 500Hz.

For this purpose, we selected 7 configurations depending on the number of required channels so as to maintain at least 1000Hz per-channel sample rate (table 3.1). Measurements of up to 6 channels (single-ended or differential alike) can use the better performing sinc5+sinc1 filter. Measurements of 7 – 10 channels cannot achieve the desired sample rate without switching to the faster yet more noisy sinc3 filter. Measurements of 11 – 16 channels cannot achieve the desired sample rate in any configuration using the AD4114, with the max per-channel sample rate degrading to $10309/16 = 644.3125\text{Hz}$ for the maximum of 16 channels.

3.2.3 Amplification stage

The chosen AD4114 ADC has $N = 24$ bits of resolution with one LSB equating to the following in the bipolar measurement mode.

$$\text{LSB} = V_{REF} / (2^{N-1} \cdot 0.1) \quad (3.1)$$

Using the internal 2.5V reference, $\text{LSB} \approx 2.98\mu\text{V}$. The noise level varies depending on the sample rate. At the single-channel $f_s = 1007\text{Hz}$, 6 channel $f_s = 6211\text{Hz}$, and all channel $f_s = 10309\text{Hz}$, the ADC noise is specified to be around $\overline{V_{na,1007}^2} = 20\mu\text{V}$, $\overline{V_{na,6211}^2} = 74\mu\text{V}$ and $\overline{V_{na,10309}^2} = 1068\mu\text{V RMS}$ respectively. Without an amplification stage, this would result in the following SNRs relative to a 1mV signal:

$$\begin{aligned}
\text{SNR}_{(\text{dB})} &= 10 \log_{10} \left(\frac{(1\text{mV})^2}{V_{na}^2} \right) \\
\text{SNR}_{1007} &\approx 34\text{dB} \\
\text{SNR}_{6211} &\approx 23\text{dB} \\
\text{SNR}_{10309} &\approx -0.6\text{dB}
\end{aligned} \tag{3.2}$$

In the case of 7+ channels, a 1mV signal slightly dips under the noise floor, becoming unmeasurable. Thus, a preamplification stage is required for enabling the use of all channels.

We choose the AD8226 [24] instrumentation amplifier as the central component of the preamplification stage. It is inexpensive, readily available, has more than sufficient CMRR ($> 80\text{dB}$) and can be configured for a gain of $G = 1$ to 1000 using a resistor $R_G = 49.4\text{k}\Omega / (G - 1)$. Its RMS voltage noise for our desired bandwidth of $\Delta f = 500\text{Hz}$ is:

$$\begin{aligned}
\sqrt{V_{ni}^2} &= \sqrt{500\text{Hz} \cdot (22\text{nV}/\sqrt{\text{Hz}})^2} = 0.492\mu\text{V} \\
\sqrt{V_{no}^2} &= \sqrt{500\text{Hz} \cdot (120\text{nV}/\sqrt{\text{Hz}})^2} = 2.683\mu\text{V}
\end{aligned} \tag{3.3}$$

With the preamp stage added before the ADC, the total noise of a sample and the theoretical SNR are:

$$\begin{aligned}
\text{SNR}_{(\text{dB})} &= 10 \log_{10} \left(\frac{(1\text{mV})^2}{V_n^2} \right) \\
(2.4) \Rightarrow \text{SNR}_{(\text{dB})} &= 10 \log_{10} \left(\frac{(1\text{mV})^2 \cdot G^2}{V_{ni}^2 \cdot G^2 + V_{no}^2 + V_{na}^2} \right)
\end{aligned} \tag{3.4}$$

As described in section 2.4.2, increasing the gain will increase SNR but with diminishing returns, and it will limit the available measurement range. With the specified values for the AD8226, we compute a handful of gain scenarios (table 3.2) and choose $G \approx 20$ as a fair middleground between good SNR and good input range, allowing for some fluctuations without saturating the sensors. We use the part's gain formula to compute a suitable gain resistor:

$$R_G = \frac{49.4\text{k}\Omega}{G - 1} = \frac{49.4\text{k}\Omega}{20 - 1} \approx 2600\Omega \tag{3.5}$$

The closest E6-series value is $R_G = 2200\Omega$ which would give a gain of $G \approx 22.45 \approx +27\text{dB}$.

Using an active element (the instrumentation amplifier) also gives us the opportunity to do some hardware filtering. To combat high frequency noise, we add a first order RC low-pass filter with a cutoff of $f_c \approx 500\text{Hz}$. Our implementation uses $R = 330\Omega$ and $C = 1\mu\text{F}$ for a theoretical 3dB cutoff frequency of:

$$f_c = \frac{1}{2\pi \cdot 330\Omega \cdot 1\mu\text{F}} = 482.28\text{Hz}$$

Table 3.2. Full scale, SNR for various gains

Gain	Approx. range	SNR
1	5V	34dB
2	2.5V	40dB
5	1V	48dB
10	500mV	54dB
20	250mV	59dB
50	100mV	64dB
100	50mV	65dB
1000	5mV	66dB
∞	0	66.16dB

To encode the bipolar differential measurement as a unipolar signal, we bias the amplifier's output voltage using the AD8226's V_{REF} input. As specified in the datasheet, this pin must be driven with less than 2Ω impedance, so just a simple resistor divider wouldn't suffice. We use a common-collector voltage buffer after a $10k\Omega - 10k\Omega$ divider to amplify the drive strength, biasing the output to approximately $V_{REF} \approx 5V/2 - 0.7V = 1.8V$.

To help with transient power supply fluctuations, the instrumentation amplifier's positive supply as well as V_{REF} are decoupled with $0.1\mu F$ capacitors. The second should also reduce BJT flicker noise.

The overall circuit diagram and a picture of the built amplifier board are presented in figure 3.1.

We validate this preamplifier board with two tests using the ADALM2000 platform. It powers the board, generates test signals, and measures the amplified output.

For the first test, a sinusoidal $A_{test} = 50mV$ peak-peak test signal is applied. The measured waveform displays the expected DC bias of $V_{amp}^- = 1.88V$, and an amplitude of $A_{amp} = 1.23V$ peak-peak, giving an effective gain of $G = A_{amp}/A_{test} = 24.6$, just within the 10% tolerance of the designed $G = 22.45$ imposed by the passive components we used. For these amplitudes, the harmonic distortion and amplifier noise are below the $-80dBFS$ noise floor of the ADALM2000.

Secondly, we test the frequency response using the Network analyzer feature of the Scopy software[25]. A $10Hz - 10kHz$ sine sweep is generated, plugged into the amplifier, and the resulting output is measured, determining both the magnitude and phase relation, which are plotted in figure 3.2. Both the measured magnitude and phase characteristics match the ideal ones well enough, especially given the 10% tolerances of the passives. For frequencies above $1kHz$, the signal amplitude decreases to small multiples of the ADALM2000 resolution, leading to increasing amounts of quantization noise, which can be seen in both the magnitude plot as an increased reported magnitude, as well as in the phase plot as phase noise. This isn't concerning, as the desired amplification and corner frequency are properly measurable and validated. These two tests successfully validate the performance of this part of the signal chain.

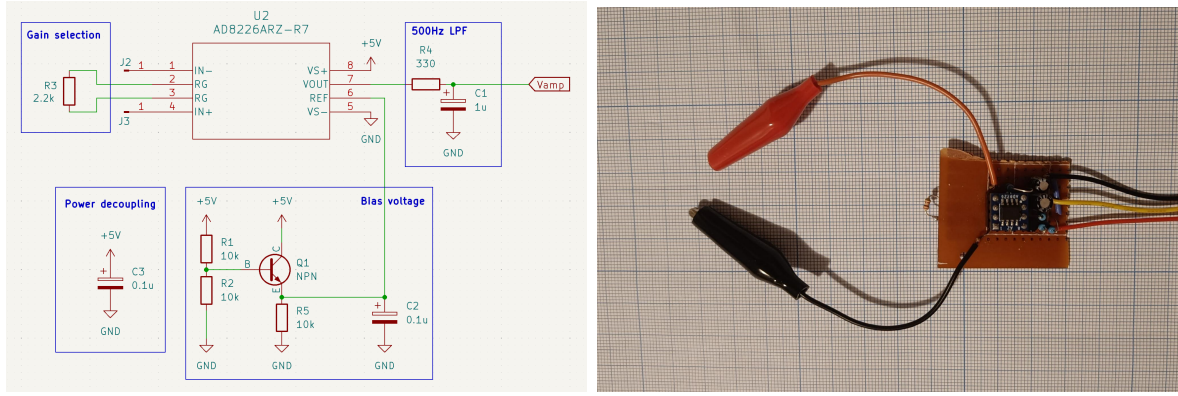


Figure 3.1. Circuit diagram and photo of built amplifier board

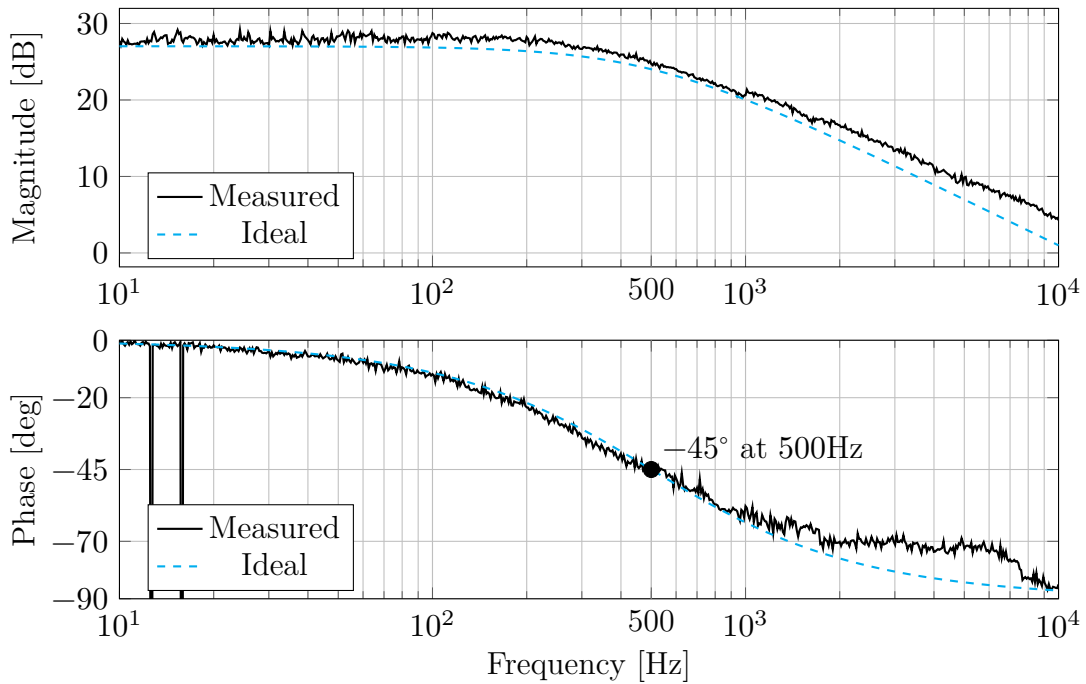


Figure 3.2. Measured preamplifier frequency response, compared to ideal component +27dB gain, 500Hz first order LPF amplifier

3.2.4 Microcontroller

For microcontroller selection, the main performance considerations are on the one hand the ability to communicate via SPI fast enough to comply with the desired sample rate, and on the other hand for it to have a fast enough clock rate and memory for the IIO Daemon which will comprise most of the final application. For further work, networking capabilities would be a nice-to-have to be able to run a standalone ROS2 node. A plethora of well known boards match these criteria, among which the very popular Espressif ESP8266 and ESP32, or the Raspberry Pi RP2040 and RP2350 would be sufficient and good alternative choices thanks to their excellent community support. The popular, yet more and more obsolete ATMEGA328p of Arduino fame isn't fast enough for both recording and further transferring all 16 channels, especially not through unmodified IIO Daemon / ros-micro implementations. We chose the Analog Devices MAX78000FTHR board thanks to its powerful specs which would also allow for some degree

of digital signal processing on the board itself. We will use the MAX78000's dedicated SPI bus which is exposed on the feather board.

TODO: MAX78000 pinout diagram?

3.3 Firmware part

TODO: Short 100-200 word intro about firmware

TODO: The no-OS framework

No-OS is a firmware framework developed by Analog Devices that defines a common API for accessing common peripherals on embedded systems. While its scope somewhat overlaps with a typical Hardware Abstraction Layer (HAL) within an Operating System, no-os focuses on providing minimal abstractions to run directly on bare metal, as the name suggests. Additionally, it provides abstractions for the platform architecture, build systems and custom vendor SDKs, such that code can be very easily ported from one platform to another.

The benefits of using such a versatile framework were constantly apparent during this project's development, which required us to change the microcontroller platform, peripherals, as well as communication protocols during prototyping. While each change required small rewrites of the codebase, the uniform structure of no-os meant we didn't have to rewrite thousands of lines of code after each hardware change.

3.3.1 Serial communication with host

In order to send data to the attached computer or carrier board, the de facto solution is using UART-over-USB, which exposes an USB interface to the attached host and UART to the microcontroller. Most commonly, microcontrollers have dedicated circuitry for UART transmit and receive, which allows the rest of the program to run concurrently to serial communication. This is implemented using two FIFO buffers, one for data pending to be sent from the microcontroller (TX FIFO) and one for received data ready to be processed by the program (RX FIFO).

De facto, most microcontroller development boards, including the MAX78000FTHR, MAX-Arduino, and the more common ATMEGA368P "Arduino-like" boards provide UART-USB translation via a dedicated IC, abstracting away the complexities of the USB communication stack.

UART modules have varying degrees of configurability of the baud rate, flow control, and framing. For most practical uses, though, including this project, one may just specify the baud rate, defaulting to 8N1 (8 data bits, no parity bit, 1 start bit, 1 stop bit) framing and no flow control. Typical baud rates include 9600 baud, a legacy of 20th century POTS modems (introduced by the V.32 ITU-T standard <empty citation>) and 115200 baud, common both in last-century modem systems, but also recently popularized by its default use in the Arduino development environment.

To achieve the required bandwidth for this project, a suitable baud rate must be chosen. Given the sample rate f_s , number of channels N , and number of bytes per channel sample d , and 8N1 framing (giving 10 baud/byte), the baud rate f_{baud} must be:

$$f_{baud} \geq f_s \cdot N \cdot d \cdot 10 = 1024 \cdot 16 \cdot 3 \cdot 10 = 491520 \text{ baud} \quad (3.6)$$

Additionally, choosing a baud rate larger than this minimum will allow for less timing concerns, especially when it comes to not overflowing the receive and transmit buffers.

Baud rate selection is not universal across devices and the firmware for each hardware platform must choose a rate that is both compatible with the hardware and that satisfies relation 3.6. In the case of the MAX78000FTHR, a rate of exactly 491520 baud can be achieved by using the IBRO (Internal Baud Rate Oscillator) of 7.3728MHz with a clock divider of 15, but [we] will use a divider of 9 for a "round" value of 819200 baud to have a good margin of error and not need such strict synchronization between the UART module and the rest of the program.

3.3.2 SPI Communication basic proof of concept (2024-05 single channel demo)

For initial validation of the concept on the AD4114, **TODO: write about the PoC: 1 channel, no amplification, full sample rate, can see both EMG and ECG on it. can do nice graphs, can diagnose my RBBB**

3.3.3 No-OS Driver

Fortunately, the no-OS ecosystem already provides drivers for both the MAX78000 platform as well as the AD4114 as part of the AD717X family of ADCs[26]. What's left is to

3.3.4 IIOd integration

Defining the structure of the IIO context is a very important step as it solidifies a mapping between the device-specific functionality on the hardware side and the application-specific functionality on the software side. Choosing a versatile collection of attributes to configure the device and channels to structure the data is no trivial task with no unique solution. The exposed attributes may encode very low level details, as in find an AD part with a very bare-metal attribute mapping, or more high level ones. Choosing which configuration is to be specified on the software side and which configuration is to be derived from that on the firmware side of *pula mea ce scriu aici... vreau sa zic ca nu o sa expun direct registri, ci o sa expun chestii mai high level: frecventa de mains hum (50/60), parametri de filtrare, poate sample rate daca o sa imi dau variante (posibil tho sa fie fix la 1k), si pe baza lor se ocupa firmware sa puna registrii cum trebuie.*

For lower-level functionality, an IIO debug interface is also specified, which allows for direct manipulation of the ADC's registers. This shouldn't be normally used, but allows for total control of the device in case the user needs it.

1. Defining the desired iio context structure: what do we expose, in which formats, etc 2. Handler functions 3. Boom done, connect to it from software

3.3.5 Continuous and asynchronous data acquisition using IIO triggers

By default, the expected behaviour of IIO devices is to start acquisition on selected channels when requested (via an OPEN command), and stop and disable the channels once the requested number of samples is reached. In low sample rate scenarios, the host may request the next acquisition window very soon after the current one finishes, and assuming the time delay between windows can be minimized with respect to the overall window size, this could prevent missing any samples and only introduce a small amount of phase drift. Additionally, there are high-speed applications in which continuity between acquisition windows is not strictly required. On the other hand, with our roughly 16kHz sample rate (cca. 60 – 62µs/sample)

and requirement to not miss any samples, this application has very little room for error in timing. As per [27], modern real-time linux setups have scheduler latencies in the tens and hundreds of microseconds, which realistically leaves no time to actually receive the current window's buffer and send a new acquisition request without missing any samples, even if the acquisition channels were left enabled.

TODO: figure, vertical stack of sequence diagrams of: normal iio polling with phase drift in slow ADC, fast ADC - loss of samples, trigger and buffer

With this general issue in mind, IIO defines triggers as software or hardware events that initiate data acquisition without requiring regular polling from the host. Data captured as a result of triggers is stored in circular buffers and provided with minimal latency whenever the host polls next.

With the AD4114 in continuous conversion mode, two trigger sources are apparent: using the $\text{DOUT}/\overline{\text{RDY}}$ behaviour in continuous conversion mode (as described in ??) or using a timer to poll the AD4114 at least as fast as the conversion rate. While the first option is more elegant and wouldn't do any unnecessary polling, it requires nonstandard operation of the SPI protocol by keeping CS active and monitoring the MISO ($\text{DOUT}/\overline{\text{RDY}}$) line, which isn't generally possible unless webitbang the SPI protocol or have access to more configurable IO options like on FPGA platforms or the Programmable Input and Output subsystem of RP2040 fame ([28]). The second option of setting up a timer interrupt which polls the AD4114 status register to check if a new conversion result is available

TODO: more :)

3.3.6 ROS2 on network-capable microcontrollers

In recent years, there has been a wave of developments in fast, network-capable, cheap microcontroller platforms, leading to products such as the Espressif ESP8266 and ESP32, or the Raspberry Pi Pico-W, which are capable of running ROS2 nodes all by themselves.

TODO: more :)

3.3.7 Digital filtering of the signal

Although the digital signal processing part of such biopotential measurement systems is very important for a full product, it is not the main topic of this thesis, and yet the chosen system architecture offloads most of the signal conditioning to the digital domain. The main source of noise is mains hum, with a fundamental frequency of 50Hz and various harmonics, which can all be removed using a notching filter, resulting in minimal loss of information in other portions of the frequency spectrum. The Quality factor Q of the filter controls how narrow the attenuated frequency bands are, and it can be adjusted. It poses a tradeoff between good noise rejection and signal quality, and is designed to be adjustable. Experimentally, we found values of around 5-10 to give the best waveforms for visual inspection. Figure X illustrates the tradeoff between small Q doing ??? and high Q doing ???.

Taking from scipy's implementation of an IIR Notching filter ('scipy.signal.iircomb'), I also use the structure described in cite: Sophocles J Orfanidis, "Introduction to Signal Processing", Prentice-Hall, 1996, ch. 11, "Digital Filter Design" thanks to its adjustable Q factor, numerical

stability even at higher orders, and default rejection of the DC component of the signal, which are all very useful for the required signal conditioning. Parametrized by the sampling frequency f_s , the cutoff frequency f_0 (which must be an integer divisor of f_s), quality factor Q , the IIR filter $H(z)$ is computed as:

$$\begin{aligned}
 N &= f_s / \omega_0 \stackrel{!}{\in} \mathbb{N} \\
 \omega'_0 &= \frac{2\pi\omega_0}{f_s} \\
 \omega_\Delta &= \frac{\omega'_0}{Q} \\
 \beta &= \tan(N * \omega_\Delta / 4) \\
 a &= \frac{1 - \beta}{1 + \beta} \\
 H(z) &= \frac{1}{1 + \beta} \cdot \frac{1 - z^{-N}}{1 - az^{-N}}
 \end{aligned} \tag{3.7}$$

TODO: check I transcribed these correctly and actually implement them myself! I only blindly copied these for now

3.4 Software part

3.4.1 ROS2 agent for non network-capable microcontrollers

In the case of microcontroller platforms that aren't network capable, including the MAX78000FTHR, the device itself won't have a means of directly connecting to the ROS2 network as a node. As such, it needs to pass along the data to another device that can. Two implementation variants are immediately evident: Using the micro-ROS framework, letting ros logic take over the serial device instead of having it communicate using the IIO protocol, and running a ROS-micro agent on the host - Using the ros-control framework and an IIO-ROS2 adapter on the host / carrier board.

Of which I chose the second because it doesn't require reconfiguring the serial port and allows for simultaneous use of ROS and IIO.

TODO: implement :)

3.4.2 libiio, pyadi-iio

ROS is for robotics, what about just doing some basic signal processing in python without setting it up as a ros node? Libiio has got you covered! Pyadi-iio adds a cherry on top, exposing the entire system as a very simple to initialize and query python object.

TODO: implement :)

4 Results and validation (3pg)

To test the performance of the system:

- ECG is more difficult and as such a greater achievement. It has smaller amplitudes and very well-defined features, requiring more care all across the signal chain. - EMG is the main goal of the project, but it is more forgiving, because: - The signal has greater amplitude (about 10x? check both sources and experimental data!!!) - The signal features are not as well defined and are usually modeled like noise for signal processing.

TODO: more :) :)

4.1 Usage for EMG

TODO: more :) :)

4.2 Usage for ECG

TODO: more :) :)

5 Conclusion (1-2pg)

We present a hardware + software solution for biopotential measurements of up to 16 single-ended or 8 differential channels, with **TODO: specs**, intended for easy and affordable maintainability, using off-the-shelf components and open source hardware and software.

5.1 Future work

While initial ambitions were high, the project fell short of all proposed objectives.

Among initial ambitions which unfortunately

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